

# Intermediate Macroeconomics: Lecture W2 (Ch.17)

ECON 311 – Intermediate Macroeconomics

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Fall 2026

# What are You Going to Learn (Ch. 17)

- How firms determine investment in physical capital

- Rather than thinking of investment in the everyday sense (e.g., stock, bond, etc), investment in macroeconomics refers to investment in the context of national income accounting
  - e.g., Roads, houses, warehouses, tools
- Today's investment influences future opportunities through the accumulation of physical capital

There are two main reasons to study physical investment more closely:

- Investment fluctuates much more than consumption
  - In year 2006, nearly 20 percent of GDP
  - In year 2009, less than 13 percent of GDP
- It is the key economic link between the present and the future
  - Investment in physical capital, human capital, and ideas are key for economic growth

# How Do Firms Make Investment Decisions

- How much should a business invest?
  - Should Amazon build a new distribution center?
  - Should Walmart open a store in downtown Toronto?
- Based on firm's profit-maximization problem (Ch.4), a business should keep investing in physical capital until

$$MPK = R$$

MPK equals the rental price of capital

- In the simple model, the rental rate of capital equals the interest rate
  - Capital comes from the decision by households to forgo consumption and save instead
  - saving = investment = supply of capital

# Arbitrage Equation

- First, imagine a person can either save or buy a pizza oven
  - The price of the oven is  $p_k$  – invest  $p_k$  or spend  $p_k$  on oven
- Saving in a bank account would yield return:

$$Rp_k$$

- Buying an oven would yield

$$MPK + \Delta p_k$$

where  $\Delta p_K = p_{K,t+1} - p_{K,t}$  – the price of the oven may go up

- Since investor can switch between the two options at anytime (and no cost), the two returns must be the same

$$Rp_k = MPK + \Delta p_K$$

# Arbitrage Equation

- Rearrange the equation:

$$\frac{MPK}{p_K} = R - \frac{\Delta p_K}{p_K}$$

- Invest in ovens until the MPK of ovens equals the difference between the interest rate and the growth rate of the price of ovens
- The left hand side tells how much of the MPK for each dollar spend on oven and it must equal to the rate of return from saving account minus the growth rate in price
- The variables on the LHS can be control by the firm but they cannot change the variables on the RHS (in a competitive market)

# The User Cost of Capital

- The growth rate in price =  $\frac{\Delta p_K}{p_K}$ 
  - When it is positive, there is capital gain
  - When it is negative, there is capital loss
- Price of capital changes because of
  - Technological change—for example, the price of computer usually declines over time
  - Scarce resources: the price of structures like factories or retail stores, in contrast, usually goes up over time as land is increasingly scarce

# The User Cost of Capital

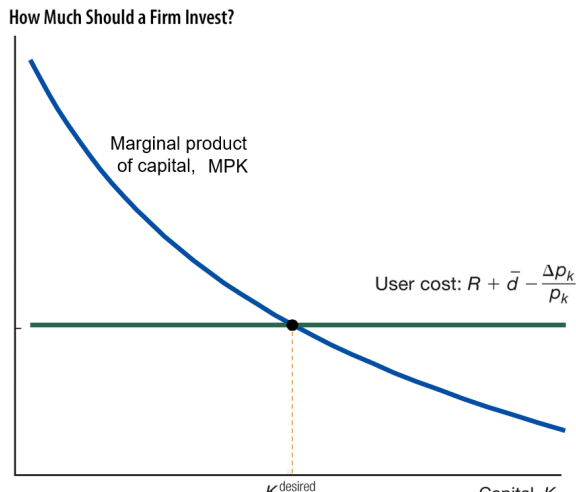
- Almost all capital would depreciate (as learn in the Solow model) which is also a cost of holding capital
- Include depreciation in the equation

$$\frac{MPK}{p_K} = R + \delta - \frac{\Delta p_K}{p_K}$$

- Firms invest until the marginal product of capital falls to equal the user cost of capital
  - User cost is the total cost to the firm of using one more unit of capital – forgone interest, depreciation and forgone capital gain
  - It is in per-dollar term

# How Much Should a Firm Invest

- The blue line shows diminishing MPK – as the  $K$  increases, the MPK falls
- The green line shows the user cost of capital is the same no matter what the level of capital



# Investment and the Corporate Income Tax

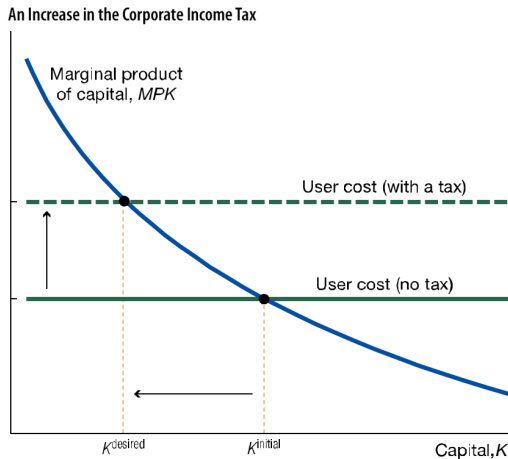
- How does corporation income tax would affect investment?
- Adding a constant tax component  $\tau$ :

$$Rp_K = (1 - \tau)MPK - p_K\delta + \Delta p_k$$

- The additional earning from investing for one unit of capital is  $MPK$ , which is  $\tau$  proportion is taxed away
- Rearrange it:
$$\frac{MPK}{p_K} = \left[ R + \delta - \frac{\Delta p_K}{p_K} \right] / (1 - \tau)$$
- Corporate income tax raises the user cost of capital

# An Increase in the Corporate Income Tax

- An increase in the corporate income tax from zero to some positive rate raises the user cost of capital
- This requires the MPK to rise, the firm desires a smaller amount of  $K$



# Case Study: Corporate Income Taxes across Countries

- Corporate tax rates vary across countries, implying differences in the user cost of capital
  - Then, it would affect the investment and the capital stock
- How large is this variation in corporate income tax?
  - 34 percent in France
  - 8.5 percent in Switzerland
- But, the implied user cost of capital varies less from country to country

# Corporate Income Tax Rates and the User Cost of Capital, 2018

- While the simple calculation shows there is positive relationship between tax rate and user cost of capital, the variation is smaller in user cost of capital
- It is because the user cost of capital is determined by other factors (e.g., depreciation rate, etc) as well

| Country     | Corporate tax rate | User cost of capital |
|-------------|--------------------|----------------------|
| France      | 34.4               | 12.0                 |
| Mexico      | 30.0               | 11.3                 |
| Korea       | 25.0               | 10.5                 |
| Italy       | 24.0               | 10.4                 |
| Japan       | 23.2               | 10.3                 |
| USA         | 21.0               | 10.0                 |
| Germany     | 15.8               | 9.4                  |
| Canada      | 15.0               | 9.3                  |
| Switzerland | 8.5                | 8.6                  |

# From Desired Capital to Investment

- Use the standard production model to pin down relationship between user cost and investment
- Using the standard production function, we can calculate the MPK

$$Y = \bar{A}K^{1/3}L^{2/3}$$

$$MPK = \frac{1}{3} \frac{Y}{K} = \frac{1}{3} \bar{A} \left( \frac{L}{K} \right)^{2/3}$$

when capital increases, the MPK falls according to the equation

- Also, we have the standard capital accumulation

$$K_{t+1} = I_t + (1 - \delta)K_t$$

# From Desired Capital to Investment

- Let  $uc = \left[ R + \delta - \frac{\Delta p_K}{p_K} \right] / (1 - \tau)$
- Use the optimal condition from the firm problem

$$uc = \frac{MPK}{p_K} = \frac{1}{p_K} \frac{1}{3} \frac{Y}{K} \implies \frac{Y}{K} = 3(uc)(P_k)$$

- Rewrite capital accumulation equation and divide both sides by  $K_t$ :

$$\frac{\Delta K_{t+1}}{K_t} = \frac{I_t}{K_t} - \delta$$

- Then, rewrite this equation and use the first equation:

$$\frac{\Delta K_{t+1}}{K_t} = \frac{I_t}{K_t} \frac{Y_t}{Y_t} - \delta = \frac{Y_t}{K_t} \frac{I_t}{Y_t} - \delta = 3(uc)(P_k) \frac{I_t}{Y_t} - \delta$$

# From Desired Capital to Investment

- Let  $g_K = \frac{\Delta K_{t+1}}{K_t}$  and rearrange the equation:

$$\frac{I_t}{Y_t} = \frac{g_K + \delta}{3(uc)(P_k)}$$

- The investment rate depends on three terms
  - The desired growth rate of capital stock ( $g_K$ )
  - The depreciation rate ( $\delta$ )
  - The user cost of capital  $\left(uc(R, \delta, \frac{\Delta P_K}{P_K}, \tau)\right)$
  - The price of capital:  $P_k$
- A higher user cost of capital leads to a lower investment rate

# From Desired Capital to Investment

Key endogenous variables in the macroeconomy:

- Growth equations in Chapters 5 and 6
  - The investment rate in Chapters 5 and 6 was a constant fraction of income.
  - This provides microfoundations for the investment rate (and the consumption share)
- Combine Euler equation (from consumption model)
  - Pins down long-run interest rate and user cost of capital

# Next Lecture

- Cover Ch.11 next lecture